Integration by Substitution

ClipSAT Printable Practice Suite (clipsat.org) • Topic 7.01

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: u-substitution

Evaluate each integral using substitution:

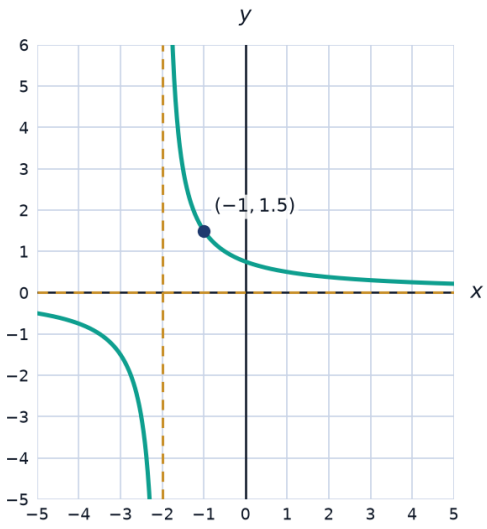
a $\int 2x(x^2+1)^5\,dx$

b $\int \frac{x}{\sqrt{x^2+9}}\,dx$

c $\int \cos(3x)\,dx$

2 [Foundational] • Target: Definite integrals with substitution

Evaluate $\int_0^2 x\sqrt{x^2+1}\,dx$ exactly, changing the limits with $u = x^2 + 1$.



3 [Foundational] • Target: Substitution with trigonometric and exponential integrands

Evaluate $\int \frac{2x+3}{x^2+3x+1}\,dx$, recognizing the numerator as the derivative of the denominator.

4 [Foundational] • Target: Choosing the substitution

For each integral, state a suitable substitution u and evaluate:

a $\int x^2(x^3-1)^7\,dx$

b $\int \tan x\,dx = \int \frac{\sin x}{\cos x}\,dx$

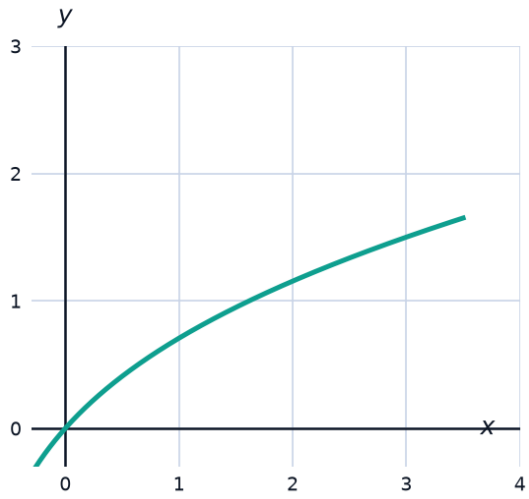
5 [Foundational] • Target: Substitution with radicals

Evaluate $\int x\sqrt{2-x}\,dx$ using the substitution $u = 2 - x$ (so $x = 2 - u$).

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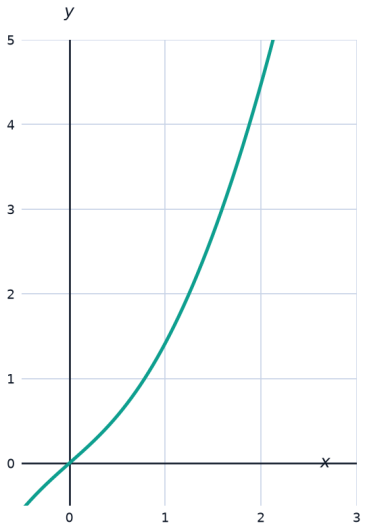
6 [Foundational] • Target: Substitution with radicals

Evaluate $\int_0^3 \frac{x}{\sqrt{x+1}}\,dx$ using $u = x + 1$ (so $x = u - 1$), changing the limits.



7 [Foundational] • Target: Visualizing an area found by substitution

The graph shows $f(x) = x\sqrt{x^2+1}$ on $[0, 2]$. Evaluate $\int_0^2 x\sqrt{x^2+1}\,dx$ exactly using $u = x^2 + 1$, and match it to the shaded area.



Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Substitution with double angle and reciprocal forms

Evaluate $\int \sin x \cos x\,dx$ two ways: (a) using $u = \sin x$, and (b) using the identity $\sin x \cos x = \frac{1}{2}\sin(2x)$. Confirm both give equivalent answers.

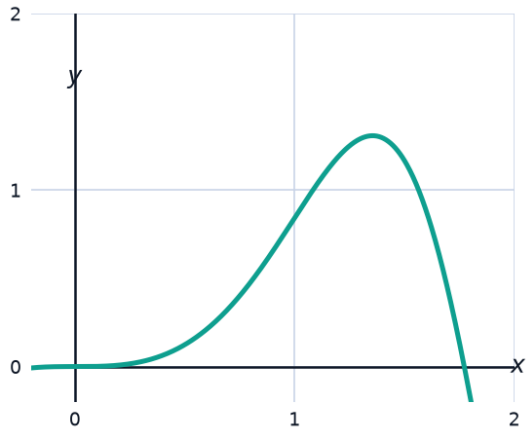
9 [Intermediate] • Target: Substitution with double angle and reciprocal forms

Evaluate $\int \frac{1}{x \ln x}\,dx$ using $u = \ln x$.

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10 **[Intermediate]** • **Target: Substitution in motion and rate problems**

A particle's velocity is $v(t) = t\sin(t^2)$ m/s. Find its displacement from $t = 0$ to $t = \sqrt{\pi}$, using $u = t^2$.



11 **[Intermediate]** • **Target: Recognizing when substitution simplifies a problem**

Evaluate $\int (x+1)\sqrt{x^2+2x+5}\,dx$ by noticing the numerator is (up to a constant) the derivative of the expression under the root.

12 **[Intermediate]** • **Target: Substitution with trigonometric and exponential integrands**

[Techniques of Integration · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

13 **[Intermediate]** • **Target: Choosing the substitution**

[Partial Derivatives & the Gradient · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

14 **[Intermediate]** • **Target: Substitution with radicals**

Use the substitution $x = 2\sin\theta$ to evaluate the exact definite integral $\int_0^1 \sqrt{4 - x^2}\,dx$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

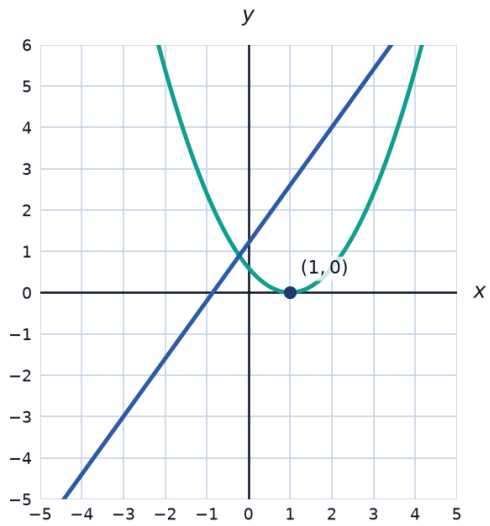
15 **[Challenging]** • **Target: Visualizing an area found by substitution** • **Cross-Lesson: 2.01 Evaluating Limits**

Level 1 · FoundationsFind the area of the region bounded by $x = y^2$ and $x = y + 2$ using integration with respect to y.

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16 [Challenging] • Target: Substitution with double angle and reciprocal forms • Cross-Lesson: 2.02 Continuity

Level 2 · MediumA 13-foot ladder leans against a vertical wall. The base of the ladder slides away from the wall at a constant rate of 2 ft/s. How fast is the top of the ladder sliding down the wall when the base is 5 feet from the wall?



17 [Challenging] • Target: Substitution in motion and rate problems • Cross-Lesson: 3.01 The Derivative from First Principles

Use the substitution $u = 2x + 1$ to find the exact value of $\int_0^4 \frac{x}{\sqrt{2x+1}}\,dx$.

18 [Challenging] • Target: Recognizing when substitution simplifies a problem • Cross-Lesson: 3.02 Differentiation Rules: Power, Product, Quotient, Chain

Use trigonometric substitution to evaluate the indefinite integral $\int \frac{1}{(4+x^2)^{3/2}}\,dx$.

19 [Challenging] • Target: u-substitution • Cross-Lesson: 4.01 Antiderivatives and Indefinite Integrals

Level 3 · AdvancedLet R be the region bounded by $y = \sqrt{x}$, $y = 2$, and the y-axis. Set up, but do not evaluate, an integral for the volume generated when R is revolved about the vertical line $x = -1$.

20 [Challenging] • Target: Definite integrals with substitution • Cross-Lesson: 4.02 Riemann Sums and Definite Integrals

Level 2 · MediumFind the angle(s) $\theta \in [0, \pi]$ at which the polar curve $r = 3\cos\theta$ passes through the origin, and determine the equation of the tangent line at that point.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: u-substitution

A ladder 13 m long leans against a wall. Base slides at 2 m/s. Rate of top falling when base is 5 m from wall:

a. $-\frac{5}{6}$ m/s

b. $-\frac{5}{12}$ m/s

c. $-\frac{2}{5}$ m/s

d. $-\frac{1}{2}$ m/s

22 [Foundational] • Target: Definite integrals with substitution

Area between $y = x$ and $y = x^2$: intersect at $x =$

a. 0 and 1

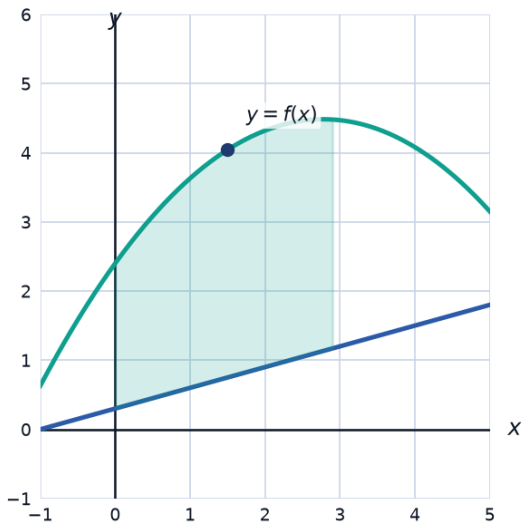
b. 1 and 2

c. 0 and 2

d. -1 and 1

23 [Foundational] • Target: Substitution with trigonometric and exponential integrands

Find the total area enclosed by the cardioid $r = 2(1 - \cos\theta)$.



a. 6π

b. 3π

c. 4π

d. 8π

24 [Foundational] • Target: Choosing the substitution

Fluid force: pressure at depth d is $\rho g d$. Force on area A:

a. $\rho g A$

b. ρA^2

c. $\rho g d A$

d. $g d$

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25 [Foundational] • Target: Substitution with radicals

Surface area of $y = \sqrt{x}$ revolved about the x-axis from 0 to 4:

a. $\frac{\pi}{3}(17\sqrt{17} - 1)$

b. $\pi(\sqrt{17} - 1)$

c. 8π

d. $\frac{\pi}{6}(17\sqrt{17} - 1)$

26 [Foundational] • Target: Visualizing an area found by substitution

Volume of solid of revolution when region $0 \le y \le e^{-x^2}$, $0 \le x \le 1$ is rotated about y-axis (shell):

a. $\pi(1 - e^{-1})$

b. $\pi(e - 1)/e$

c. $\pi/2$

d. $2\pi\int_0^1 xe^{-x^2}dx$

27 [Foundational] • Target: Substitution with double angle and reciprocal forms

Surface area of Gabriel's horn $y = 1/x$, $x \ge 1$, rotated about x-axis:

a. diverges

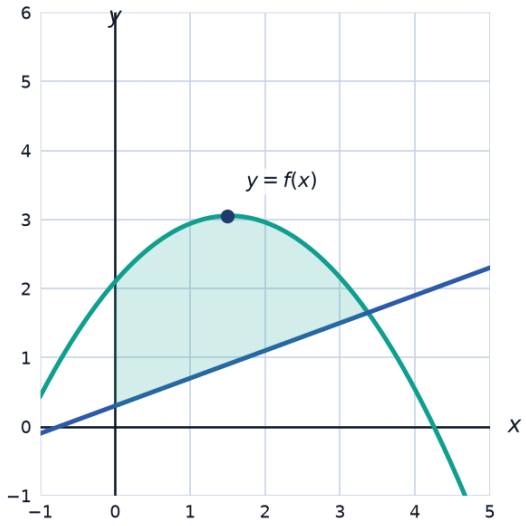
b. π

c. 2π

d. finite

28 [Foundational] • Target: Substitution in motion and rate problems

Area enclosed by $r = \sin 2\theta$ (four-petal rose):



a. $\pi/4$

b. $\pi/2$

c. π

d. 2π

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Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 **[Intermediate]** • **Target: Recognizing when substitution simplifies a problem**

Exchange integration order: $\int_0^4 \int_{\sqrt{x}}^2 f(x,y)\,dy\,dx =$

a. $\int_0^4 \int_0^y f\,dx\,dy$

b. $\int_0^2 \int_0^y f\,dx\,dy$

c. $\int_0^2 \int_y^4 f\,dx\,dy$

d. $\int_0^2 \int_0^{y^2} f(x,y)\,dx\,dy$

30 **[Intermediate]** • **Target: u-substitution**

Region bounded by $y = 3 - x^2$ and $y = -1$. Area:

a. 8

b. 16/3

c. 4

d. 32/3

31 **[Intermediate]** • **Target: Definite integrals with substitution**

Find the area of the surface formed by revolving $y = 4x$, $0 \le x \le 2$, about the x-axis.

a. $8\sqrt{17}\,\pi$

b. 16π

c. $\frac{128\pi}{3}$

d. $16\sqrt{17}\,\pi$

32 **[Intermediate]** • **Target: Substitution with trigonometric and exponential integrands**

Area of the region enclosed by the lemniscate $r^2 = 4\cos 2\theta$:

a. 4

b. 2

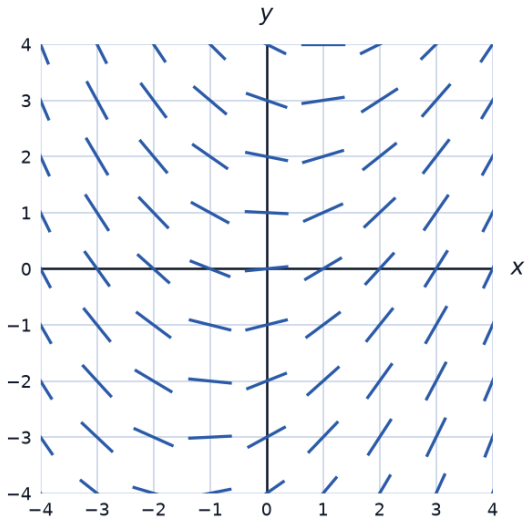
c. 8

d. π

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33 [Intermediate] • Target: Choosing the substitution

The equation $y' = ky$ models exponential growth when:



a. $k = 0$

b. $k < 0$

c. $k = 1$

d. $k > 0$

34 [Intermediate] • Target: Substitution with radicals

Integration by parts: $\int xe^x\,dx =$

a. $e^x(x-1)+C$

b. $e^x(x+1)+C$

c. $xe^x + C$

d. $\frac{x^2e^x}{2} + C$

35 [Intermediate] • Target: Visualizing an area found by substitution

Area of surface $z = \sqrt{4 - x^2 - y^2}$ over disk $x^2 + y^2 \le 4$:

a. 4π

b. π

c. 8π

d. 2π

36 [Intermediate] • Target: Substitution with double angle and reciprocal forms

Area between $y = x^2$ and $y = x$ equals:

a. 1/3

b. 1/6

c. 1/2

d. 1

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37 [Intermediate] • Target: Substitution in motion and rate problems

MVT for double integrals: $\iint_D f\,dA = f(c, d) \cdot A(D)$ requires f to be:

a. continuous on a connected region

b. positive

c. analytic

d. harmonic

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Recognizing when substitution simplifies a problem • Cross-Lesson: 3.02 Differentiation Rules: Power, Product, Quotient, Chain

Improper double integral: $\int_0^{\infty}\int_0^{\infty} e^{-(x+y)}dx\,dy =$

a. 1

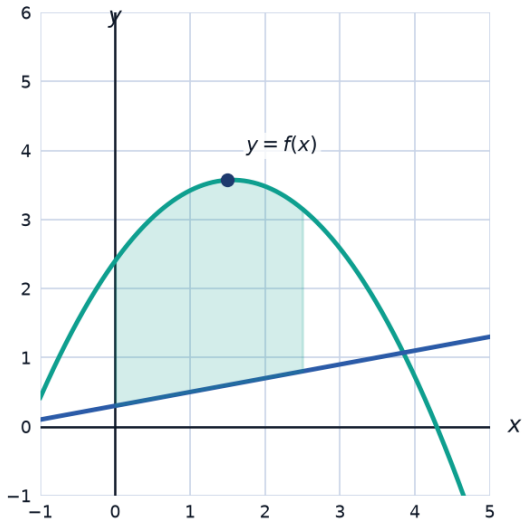
b. 0

c. ∞

d. e

39 [Challenging] • Target: u-substitution • Cross-Lesson: 4.01 Antiderivatives and Indefinite Integrals

For $f \geq 0$, the double integral represents:



a. volume below $z = 0$

b. volume between the surface $z = f(x, y)$ and the xy-plane

c. area of the surface

d. perimeter

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40 [Challenging] • Target: Definite integrals with substitution • Cross-Lesson: 4.02 Riemann Sums and Definite Integrals

Fubini's theorem allows switching the order of integration for:

a. all functions

b. only constant functions

c. non-negative or absolutely integrable functions

d. periodic functions

41 [Challenging] • Target: Substitution with trigonometric and exponential integrands • Cross-Lesson: 5.01 The Fundamental Theorem of Calculus

Green's theorem area formula for ellipse $x = a\cos t, y = b\sin t$:

a. ab

b. $2\pi ab$

c. $\pi(a+b)$

d. πab

42 [Challenging] • Target: Choosing the substitution • Cross-Lesson: 6.01 Related Rates

The region between $y = e^x$ and $y = 1$ from $x = 0$ to $x = 1$ has area:

a. 1

b. $e - 1$

c. e

d. $e - 2$

43 [Challenging] • Target: Substitution with radicals • Cross-Lesson: 6.02 Optimization and Curve Sketching

Polar tangent: the curve passes through origin when $r = 0$. Tangent angle $\psi =$

a. π

b. θ_0 where $r(\theta_0) = 0$

c. 0

d. $\pi/2$

44 [Challenging] • Target: Visualizing an area found by substitution • Cross-Lesson: 7.02 Integration by Parts

Separable double integral: if $f(x,y) = g(x)h(y)$, then $\int_a^b \int_c^d f\,dy\,dx =$

a. $\int_a^b g \int_c^d h$

b. $\int_a^b gh\,dx$

c. $\int_a^b g + \int_c^d h$

d. $(\int_a^b g)(\int_c^d h)$

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45 [Challenging] • Target: Substitution with double angle and reciprocal forms • Cross-Lesson: 8.01 Area Between Curves

Area between $y = f(x)$ and x-axis on $[a, b]$ equals:

a. $\int_a^b |f(x)|\,dx$

b. $\int_a^b f(x)\,dx$ only when $f \geq 0$

c. $f(b) - f(a)$

d. $\int_a^b f(x)^2\,dx$

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